

Class 11: AlphaFold

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Table of contents

Predicted Alignment Error for domains	9
Residue conservation from alignment file	12

```
library(bio3d)
pdb <- read.pdb("hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_4_seed_000.pdb")
```

Make a vector of input PDB file names that we can read into R

```
pdb_files <- list.files("hivpr_dimer_23119/",
                        pattern = ".pdb",
                        full.names = TRUE)

basename(pdb_files)
```

```
[1] "hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)

pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
```

hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_0
hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_0
.....

Extracting sequences

pdb/seq: 1 name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_1_seed_0
pdb/seq: 2 name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_4_seed_0
pdb/seq: 3 name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0
pdb/seq: 4 name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_0
pdb/seq: 5 name: hivpr_dimer_23119//hivpr_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_0

pdbs

```

1                                     .               .               .               .               50
[Truncated_Name:1]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGI
[Truncated_Name:2]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGI
[Truncated_Name:3]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGI
[Truncated_Name:4]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGI
[Truncated_Name:5]hivpr_dime PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGI
*****
1                                     .               .               .               .               50

51                                     .               .               .               .               100
[Truncated_Name:1]hivpr_dime GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivpr_dime GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivpr_dime GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivpr_dime GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivpr_dime GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
*****
51                                     .               .               .               .               100

101                                    .               .               .               .               150
[Truncated_Name:1]hivpr_dime QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:2]hivpr_dime QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:3]hivpr_dime QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:4]hivpr_dime QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
[Truncated_Name:5]hivpr_dime QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWPKMIGGIG
*****
101                                    .               .               .               .               150

```

```

                                151          .          .          .          .          198
[Truncated_Name:1]hivpr_dime  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivpr_dime  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivpr_dime  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivpr_dime  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivpr_dime  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                                *****
                                151          .          .          .          .          198

```

Call:

```
pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

```
5 sequence rows; 198 position columns (198 non-gap, 0 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```
rd <- rmsd(pdb, fit=T)
```

Warning in rmsd(pdb, fit = T): No indices provided, using the 198 non NA positions

```
range(rd)
```

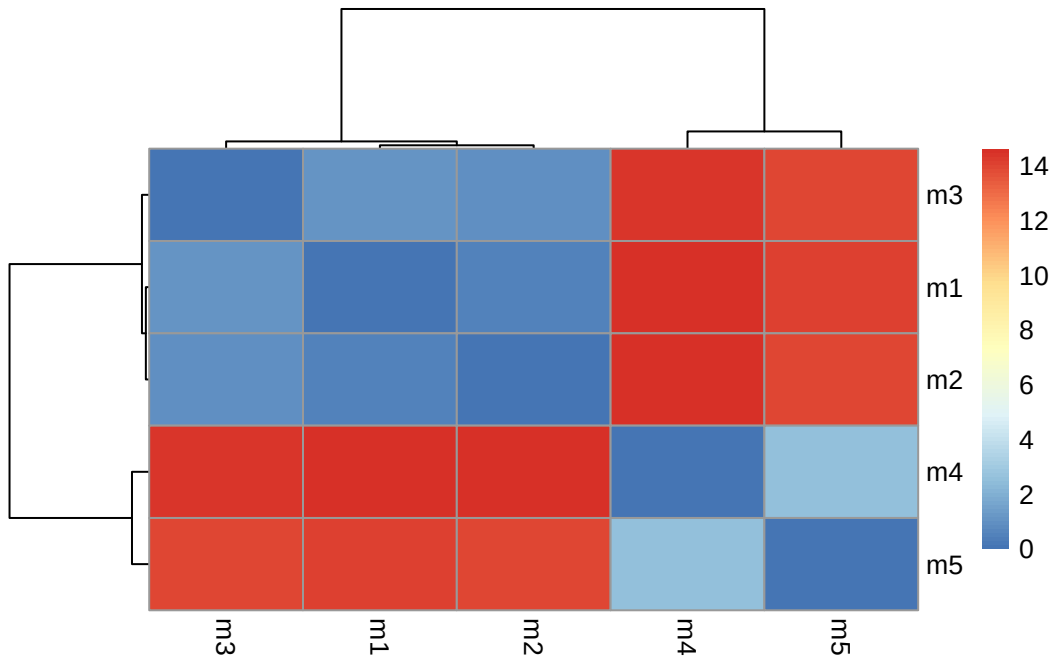
```
[1] 0.000 14.628
```

```
library(pheatmap)
```

```
colnames(rd) <- paste0("m",1:5)
```

```
rownames(rd) <- paste0("m",1:5)
```

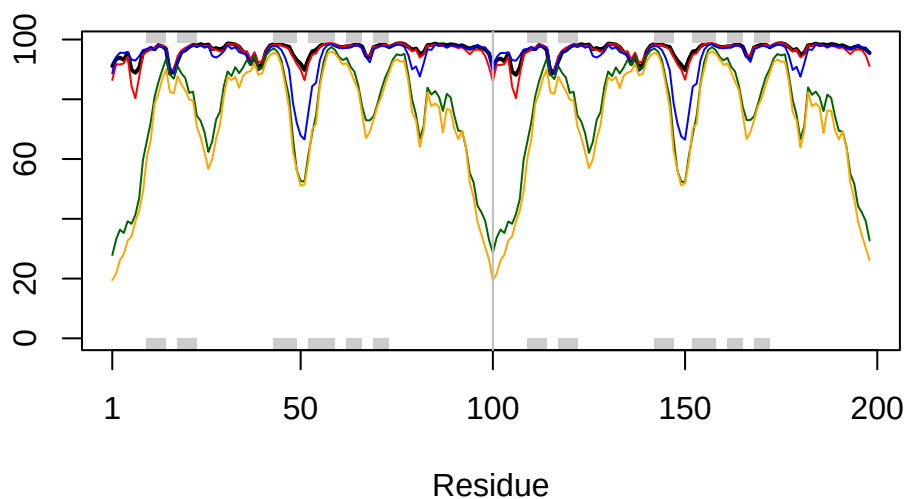
```
pheatmap(rd)
```



```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
plotb3(pdb$b[1,], typ="l", lwd=2, sse=pdb)
points(pdb$b[2,], typ="l", col="red")
points(pdb$b[3,], typ="l", col="blue")
points(pdb$b[4,], typ="l", col="darkgreen")
points(pdb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



```
core <- core.find(pdb)
```

```
core size 197 of 198 vol = 8788.529
core size 196 of 198 vol = 3258.252
core size 195 of 198 vol = 1641.096
core size 194 of 198 vol = 1498.73
core size 193 of 198 vol = 1369.427
core size 192 of 198 vol = 1313.823
core size 191 of 198 vol = 1228.996
core size 190 of 198 vol = 1186.704
core size 189 of 198 vol = 1146.395
core size 188 of 198 vol = 1102.479
core size 187 of 198 vol = 1065.499
core size 186 of 198 vol = 1024.388
core size 185 of 198 vol = 990.444
core size 184 of 198 vol = 959.404
core size 183 of 198 vol = 896.555
core size 182 of 198 vol = 862.291
core size 181 of 198 vol = 813.796
core size 180 of 198 vol = 778.382
core size 179 of 198 vol = 752.381
core size 178 of 198 vol = 727.481
```

core size 177 of 198	vol = 701.894
core size 176 of 198	vol = 665.126
core size 175 of 198	vol = 645.103
core size 174 of 198	vol = 611.674
core size 173 of 198	vol = 592.081
core size 172 of 198	vol = 574.373
core size 171 of 198	vol = 557.314
core size 170 of 198	vol = 540.852
core size 169 of 198	vol = 521.716
core size 168 of 198	vol = 504.275
core size 167 of 198	vol = 486.79
core size 166 of 198	vol = 467.6
core size 165 of 198	vol = 453.621
core size 164 of 198	vol = 438.703
core size 163 of 198	vol = 416.355
core size 162 of 198	vol = 403.203
core size 161 of 198	vol = 389.116
core size 160 of 198	vol = 380.198
core size 159 of 198	vol = 368.51
core size 158 of 198	vol = 354.354
core size 157 of 198	vol = 345.969
core size 156 of 198	vol = 333.055
core size 155 of 198	vol = 323.291
core size 154 of 198	vol = 310.454
core size 153 of 198	vol = 298.867
core size 152 of 198	vol = 287.369
core size 151 of 198	vol = 274.702
core size 150 of 198	vol = 265.237
core size 149 of 198	vol = 253.87
core size 148 of 198	vol = 242.239
core size 147 of 198	vol = 231.934
core size 146 of 198	vol = 221.935
core size 145 of 198	vol = 212.751
core size 144 of 198	vol = 202.011
core size 143 of 198	vol = 190.404
core size 142 of 198	vol = 178.064
core size 141 of 198	vol = 166.007
core size 140 of 198	vol = 154.581
core size 139 of 198	vol = 148.005
core size 138 of 198	vol = 142.521
core size 137 of 198	vol = 137.976
core size 136 of 198	vol = 133.424
core size 135 of 198	vol = 127.466

core size 134 of 198	vol = 122.719
core size 133 of 198	vol = 116.921
core size 132 of 198	vol = 111.806
core size 131 of 198	vol = 107.066
core size 130 of 198	vol = 102.136
core size 129 of 198	vol = 95.713
core size 128 of 198	vol = 91.571
core size 127 of 198	vol = 87.169
core size 126 of 198	vol = 83.202
core size 125 of 198	vol = 79.384
core size 124 of 198	vol = 75.619
core size 123 of 198	vol = 72.22
core size 122 of 198	vol = 68.808
core size 121 of 198	vol = 65.053
core size 120 of 198	vol = 59.938
core size 119 of 198	vol = 56.462
core size 118 of 198	vol = 50.965
core size 117 of 198	vol = 47.711
core size 116 of 198	vol = 44.559
core size 115 of 198	vol = 42.227
core size 114 of 198	vol = 38.533
core size 113 of 198	vol = 35.532
core size 112 of 198	vol = 32.536
core size 111 of 198	vol = 30.047
core size 110 of 198	vol = 27.602
core size 109 of 198	vol = 25.46
core size 108 of 198	vol = 23.828
core size 107 of 198	vol = 21.844
core size 106 of 198	vol = 21.094
core size 105 of 198	vol = 18.919
core size 104 of 198	vol = 18.031
core size 103 of 198	vol = 16.952
core size 102 of 198	vol = 15.508
core size 101 of 198	vol = 13.766
core size 100 of 198	vol = 13.412
core size 99 of 198	vol = 12.055
core size 98 of 198	vol = 10.539
core size 97 of 198	vol = 9.331
core size 96 of 198	vol = 7.81
core size 95 of 198	vol = 6.417
core size 94 of 198	vol = 5.507
core size 93 of 198	vol = 4.463
core size 92 of 198	vol = 3.688

```

core size 91 of 198  vol = 2.703
core size 90 of 198  vol = 2.106
core size 89 of 198  vol = 1.512
core size 88 of 198  vol = 1.06
core size 87 of 198  vol = 0.777
core size 86 of 198  vol = 0.641
core size 85 of 198  vol = 0.524
core size 84 of 198  vol = 0.473
FINISHED: Min vol ( 0.5 ) reached

```

```
core.inds <- print(core, vol=0.5)
```

```

# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     8  48     41
2    52  95     44

```

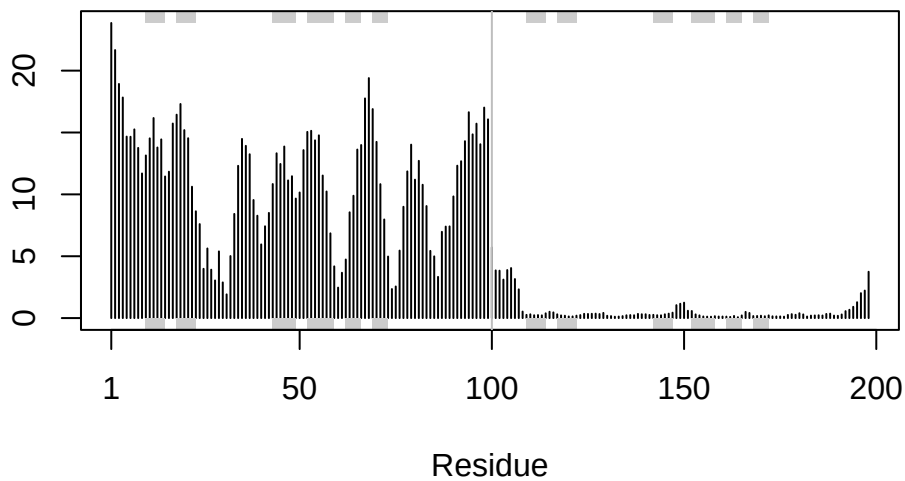
```
xyz <- pdbfit(pdb, core.inds, outpath="corefit_structures")
```

```
rf <- rmsf(xyz)
```

```

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")

```



Predicted Alignment Error for domains

```
library(jsonlite)
```

```
# Listing of all PAE JSON files
```

```
pae_files <- list.files(path="hivpr_dimer_23119/",  
                        pattern=".*model.*\\.json",  
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
```

```
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)
```

```
attributes(pae1)
```

```
$names
```

```
[1] "plddt"    "max_pae" "pae"      "ptm"      "iptm"
```

```
head(pae1$plddt)
```

```
[1] 91.12 93.50 93.94 93.19 95.50 89.81
```

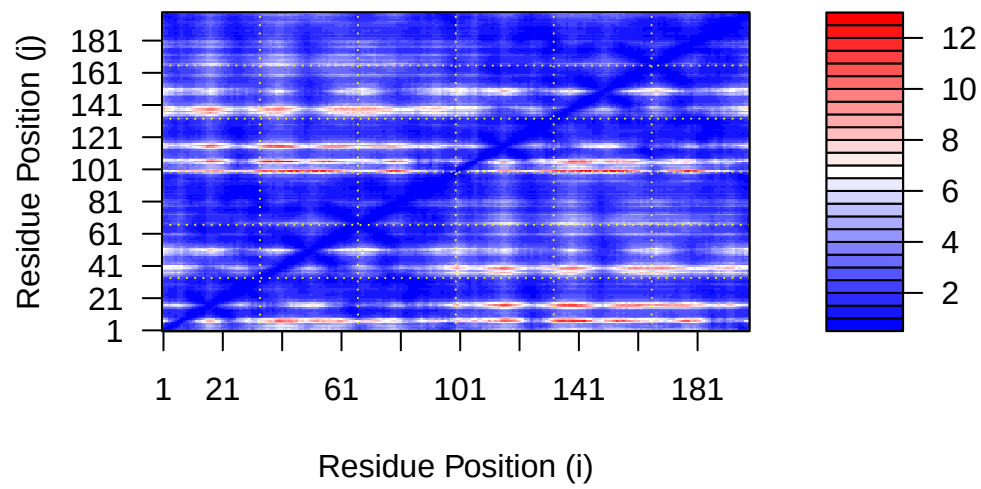
```
pae1$max_pae
```

```
[1] 12.70312
```

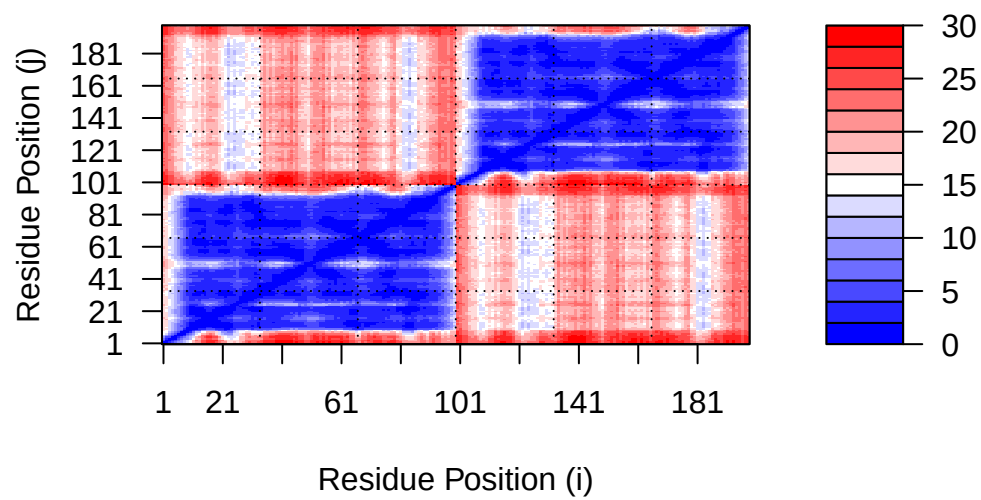
```
pae5$max_pae
```

```
[1] 29.76562
```

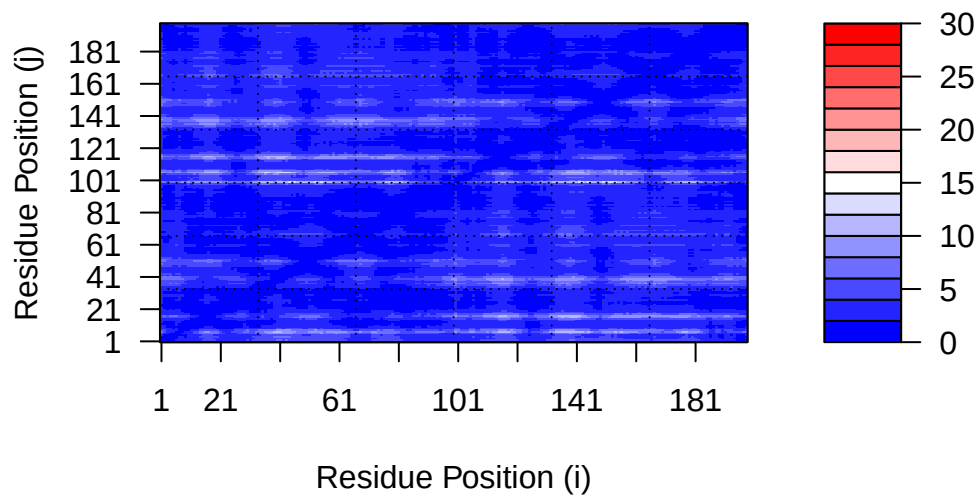
```
plot.dmat(pae1$pae,  
          xlab="Residue Position (i)",  
          ylab="Residue Position (j)")
```



```
plot.dmat(pae5$pae,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



```
plot.dmat(pae1$pae,
  xlab="Residue Position (i)",
  ylab="Residue Position (j)",
  grid.col = "black",
  zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path="hivpr_dimer_23119/",
                      pattern=".a3m$",
                      full.names = TRUE)
aln_file
```

```
[1] "hivpr_dimer_23119//hivpr_dimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

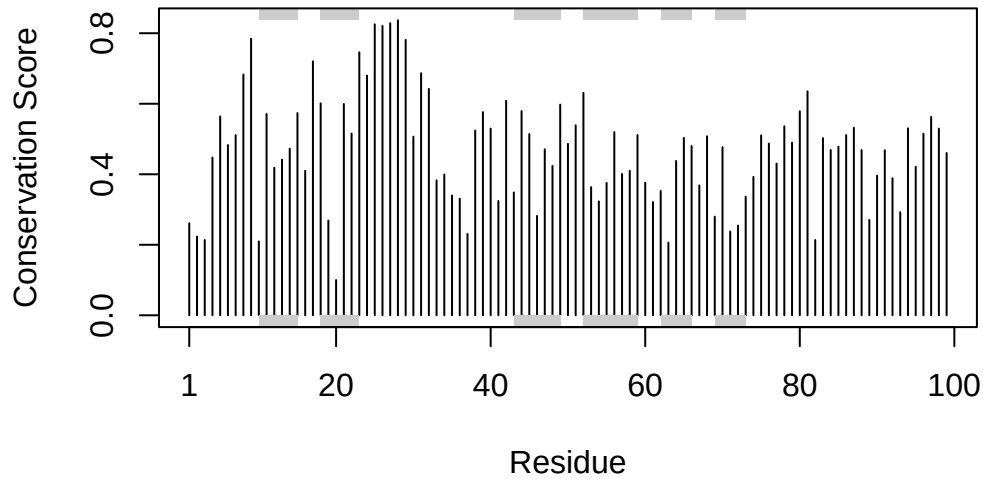
```
[1] " ** Duplicated sequence id's: 101 **"
```

```
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397 132
```

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```